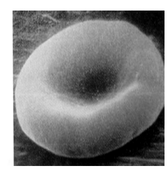
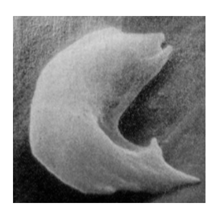
Sickle cell hemoglobin and normal hemoglobin differ in only a single amino acid out of more than 100 amino acids in the complete hemoglobin protein. This difference in a single amino acid results in the different properties of sickle cell hemoglobin compared to normal hemoglobin.

Hemoglobin is carried inside red blood cells. Normal hemoglobin dissolves in the watery cytosol of red blood cells. Sickle cell hemoglobin is less soluble in the cytosol because:

- Valine (Val) is much less water-soluble than glutamic acid (Glu).
- Amino acid 6 is in a crucial location on the outer surface of the hemoglobin protein.

The chart on the next page shows how the lower solubility of sickle cell hemoglobin results in the symptoms of sickle cell anemia.

Genes in DNA	→	**Protein**	→	**Characteristics**
2 copies of the allele that codes for normal hemoglobin (**SS**)	→	Normal hemoglobin dissolves in the cytosol of red blood cells. 	→	Disk-shaped red blood cells can squeeze through the smallest blood vessels → normal health 
2 copies of the allele that codes for sickle cell hemoglobin (**ss**)	→	Sickle cell hemoglobin can clump in long rods in red blood cells. 	→	If sickle cell hemoglobin clumps in long rods → sickle-shaped red blood cells → clogged small blood vessels + fragile red blood cells → pain, damage to body organs + anemia = sickle cell anemia 

29a. Circle the arrows in the chart that represent transcription + translation.